

Browser (client)

Camera
(video frames)

MediaPipe Hands:
21 landmarks (x, y, z)
per frame

Landmark sequence
JSON (never video)

Backend

Gates: ≥ 15 frames,
 $\geq 60\%$ presence,
 ≥ 10 fps, 0.5–10 s

Isotropic coords;
mirror to user view

Gap filling + Gaussian
smoothing ($\sigma = 60$ ms)

Idle trimming
(palm-speed threshold)

Z validity: right,
down-left, right
(gate only)

Resample to
64 points (time)

13 weighted features
(Fig. 6)

Keyed orthonormal
 Q_v (13×13)

DTW (band 0.2) $\times$
rotations $\{0, \pm 6, \pm 12\}^\circ$
vs S_1, S_2, S_3

Median of 3
 ≤ 0.216 (dev.)
 $\rightarrow$ match

Gate or Z-validity failure $\rightarrow$ capture failure (retry), never a mismatch.

Enrollment runs the same pipeline on 3 samples, stores them separately (keyed), and refuses a sample unlike both others.